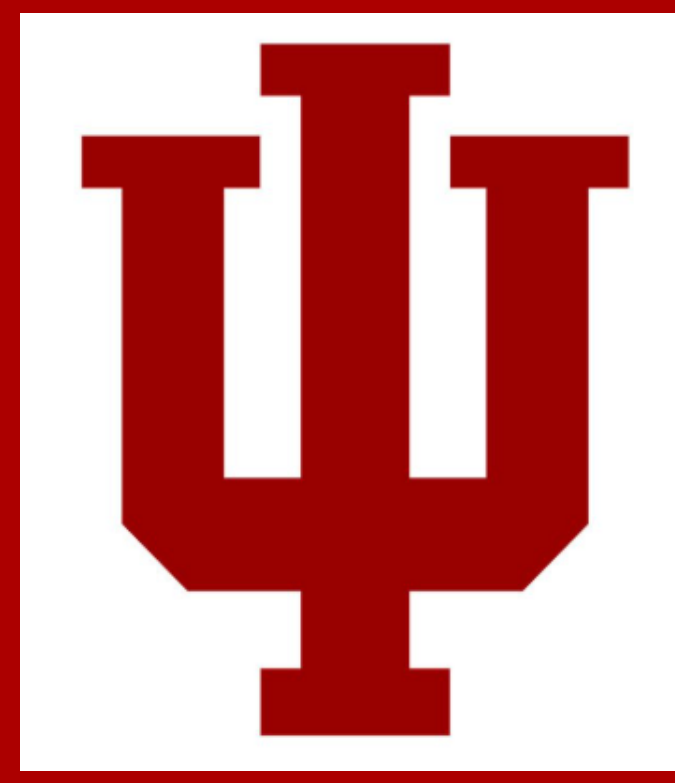




Data-Driven Analysis of Content Performance and Audience Engagement for Waste Hero

In collaboration with Yunus Thailand



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INTRODUCTION

Waste Hero, supported by **Yunus Thailand**, is a global initiative that promotes sustainability education through digital content across platforms such as Facebook, Instagram, LinkedIn, and YouTube. They focus on social business solutions to address environmental and social challenges, with Waste Hero serving as a key initiative to drive awareness around recycling and zero waste.



OBJECTIVE

To identify how content characteristics (theme and formats) and audience characteristics (age, demographics and traffic sources) influence engagement and high-intent user actions, and to derive data-driven insights and strategies that can help Waste Hero optimize its digital presence, improve conversion outcomes, and expand its sustainability and impact.

METHOD

Data Sources: Social media data from **Facebook, Instagram, LinkedIn, and YouTube**, combined with **Google Analytics** website data to capture both content performance and audience behavior.

Content Analysis : Posts were analyzed using a **text-based keyword classification approach** to assign one of six themes: Education/Learning, Youth/Community, Recycling, Circular Economy, Zero Waste, and Events/Campaigns. Content formats were standardized (video, image, carousel, reels, text) to enable cross-platform comparison.

Audience Analysis : User behavior was analyzed across **age groups, countries, and traffic sources**. Traffic sources were grouped into categories such as Google, Social (LinkedIn, Facebook, Instagram, YouTube), Email/Direct, Campaign, and Referral.

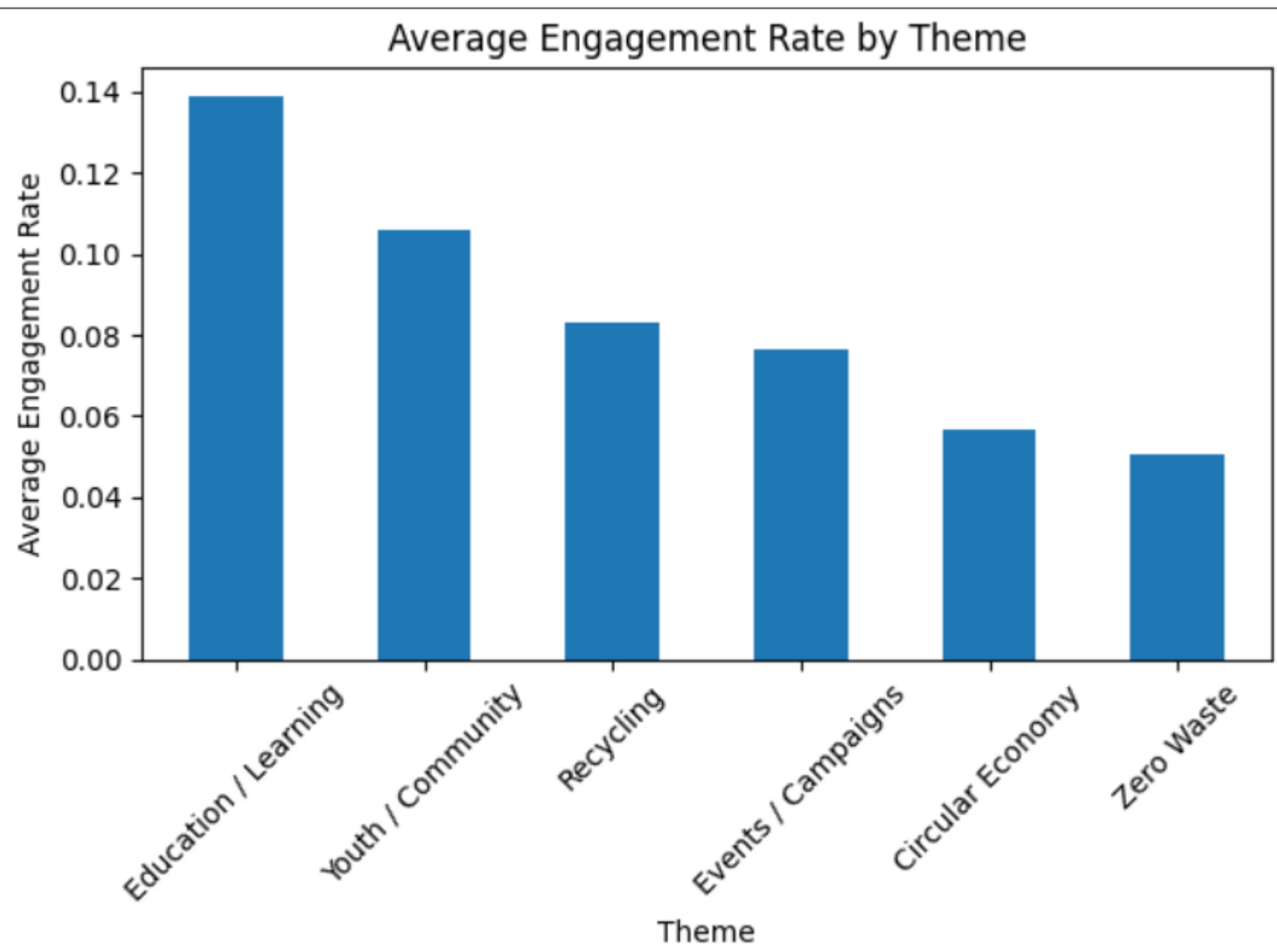
Metrics: **Engagement Rate** = interactions / views

Download Rate = downloads / views

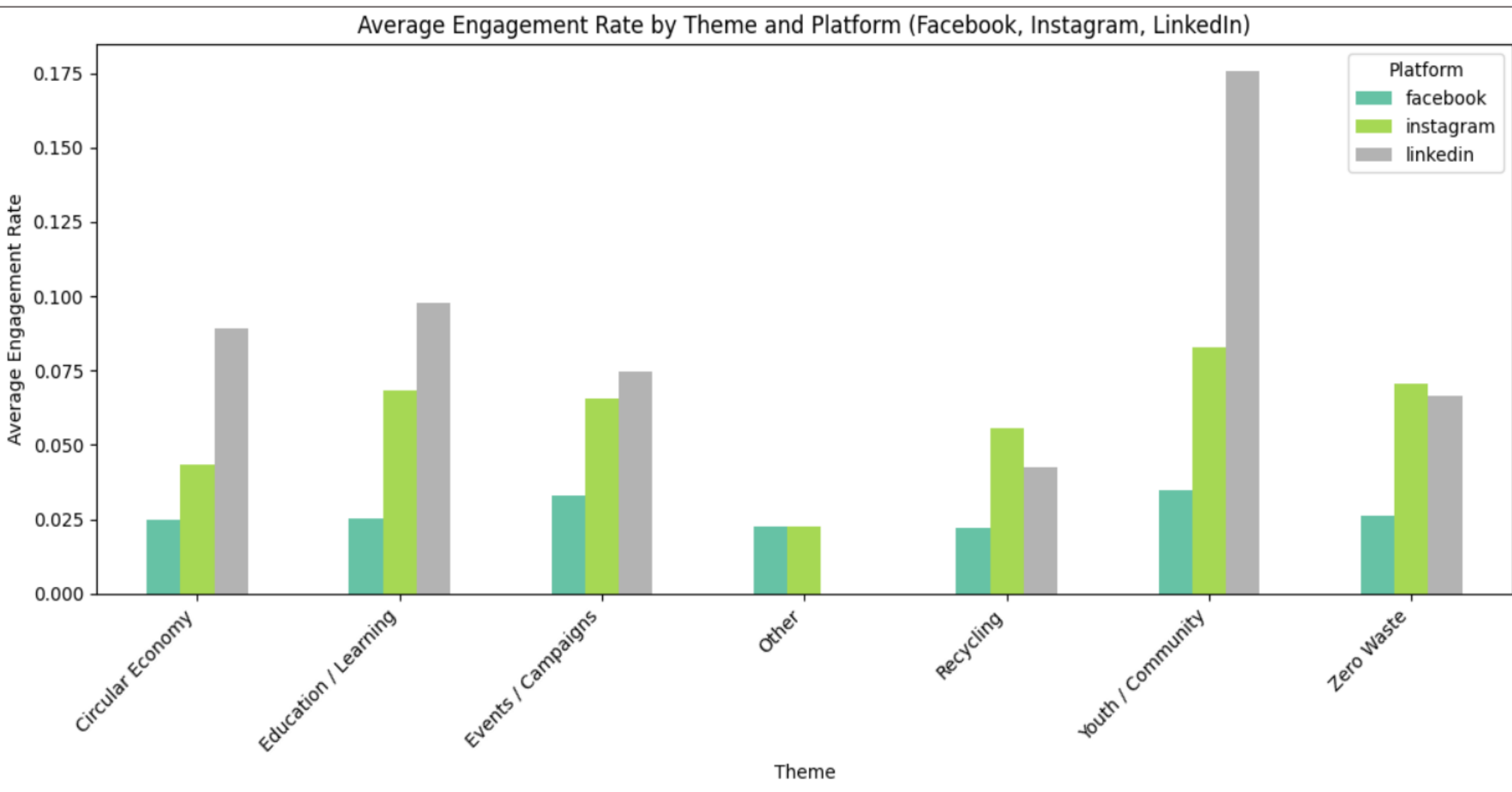
These normalized metrics allow comparison across platforms and audience segments with different levels of visibility.

RESULTS

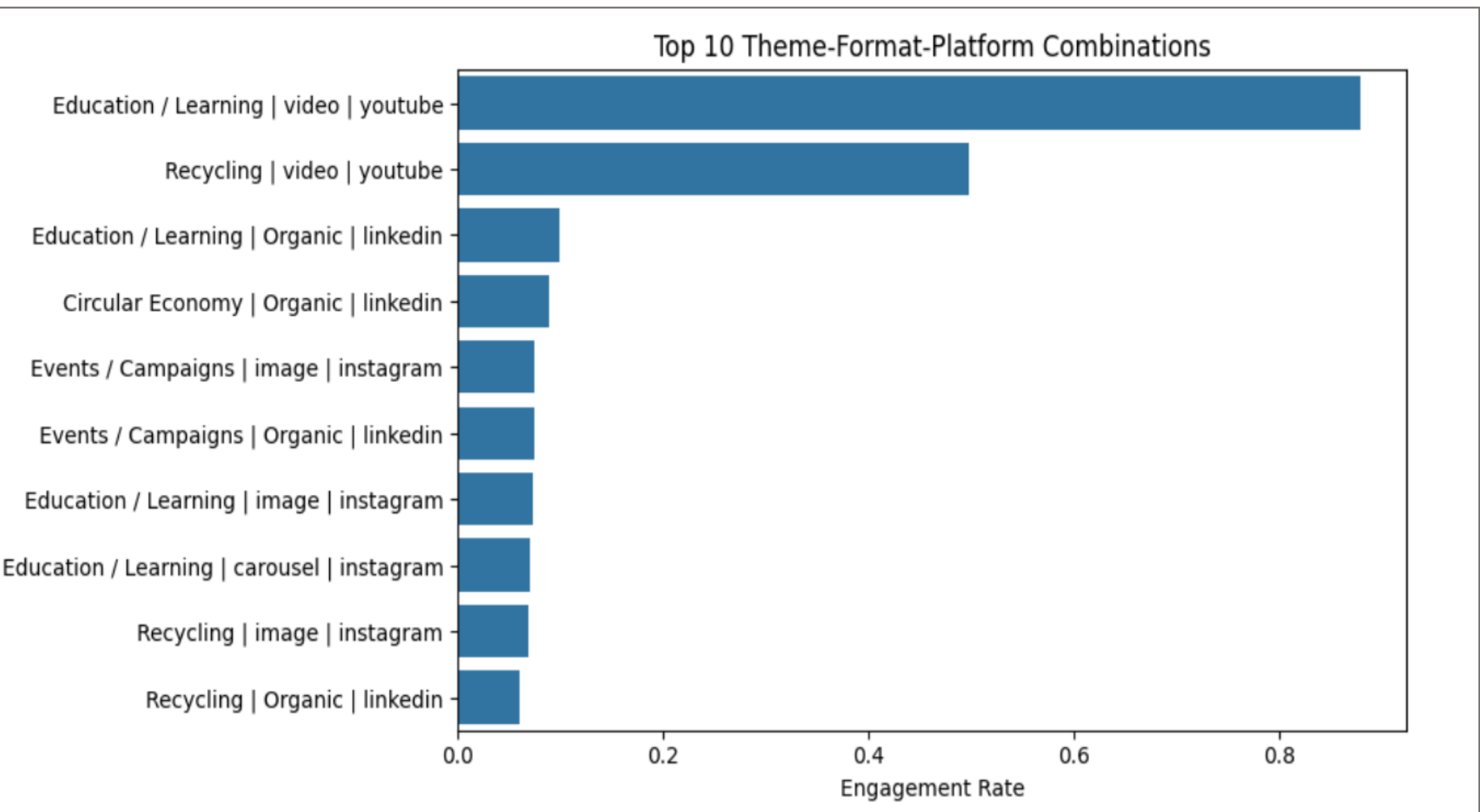
Content Performance (Theme × Format × Platform)



Education/Learning shows strongest performance, followed by Youth/ Community. Zero Waste and Circular Economy exhibit lower engagement.

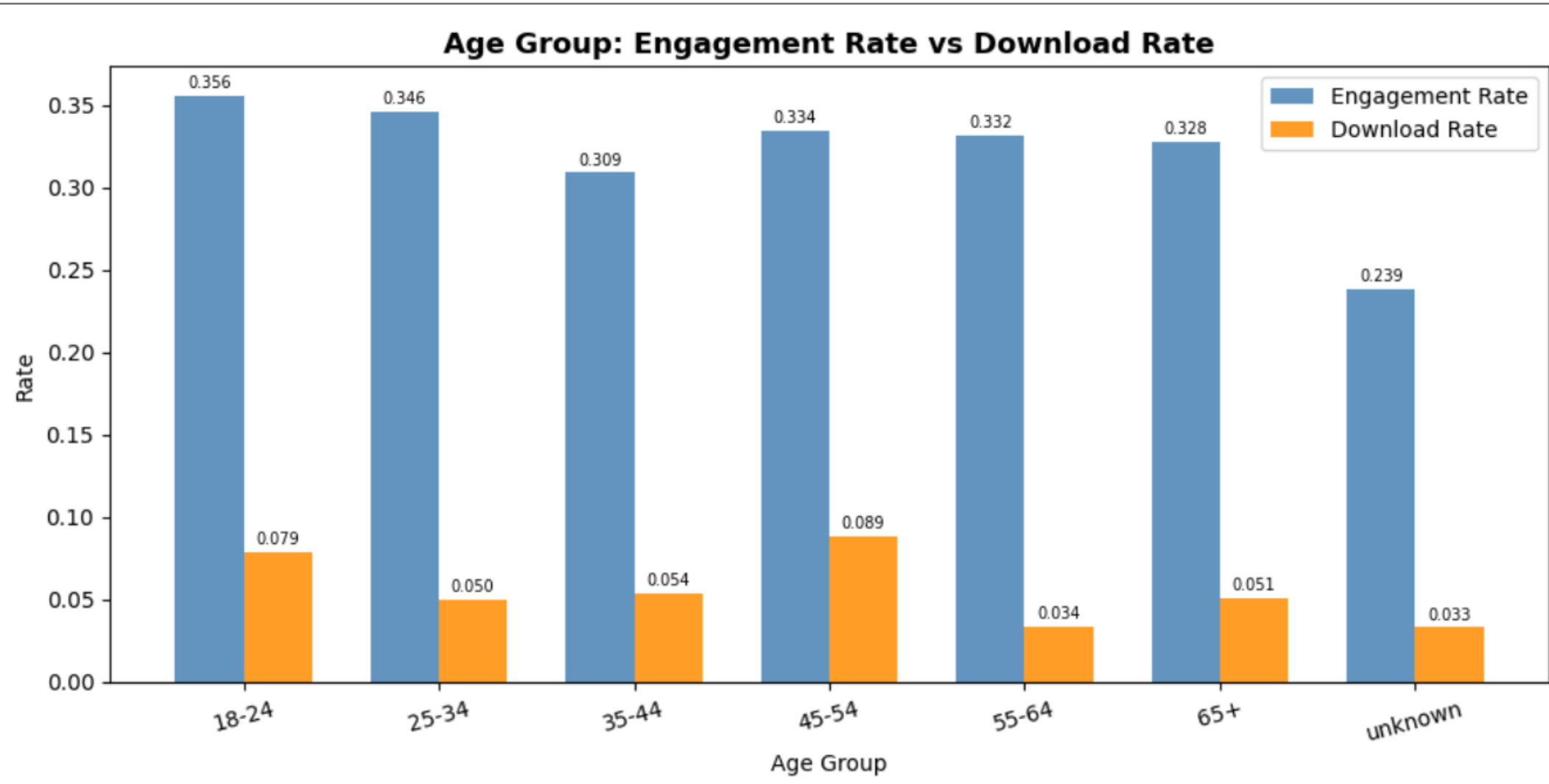


Engagement varies by platform. LinkedIn shows the highest interaction for Education, Community themes, while Instagram performs moderately across themes. Facebook consistently shows lower engagement.

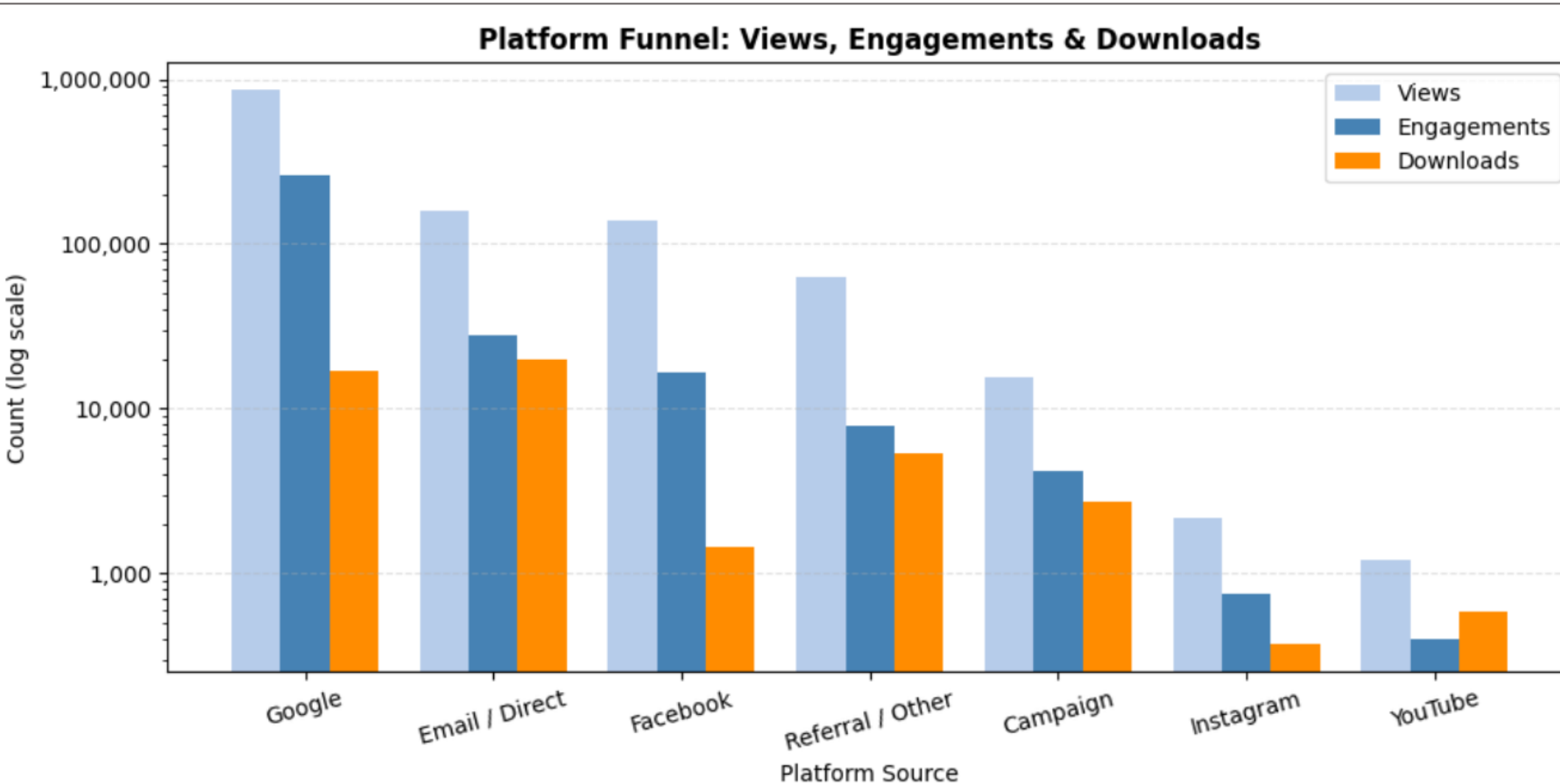


Highest-performing combinations are driven by alignment of theme, format, and platform. Video content dominates on YouTube, LinkedIn favors organic educational posts and Instagram performs well with image and carousel formats.

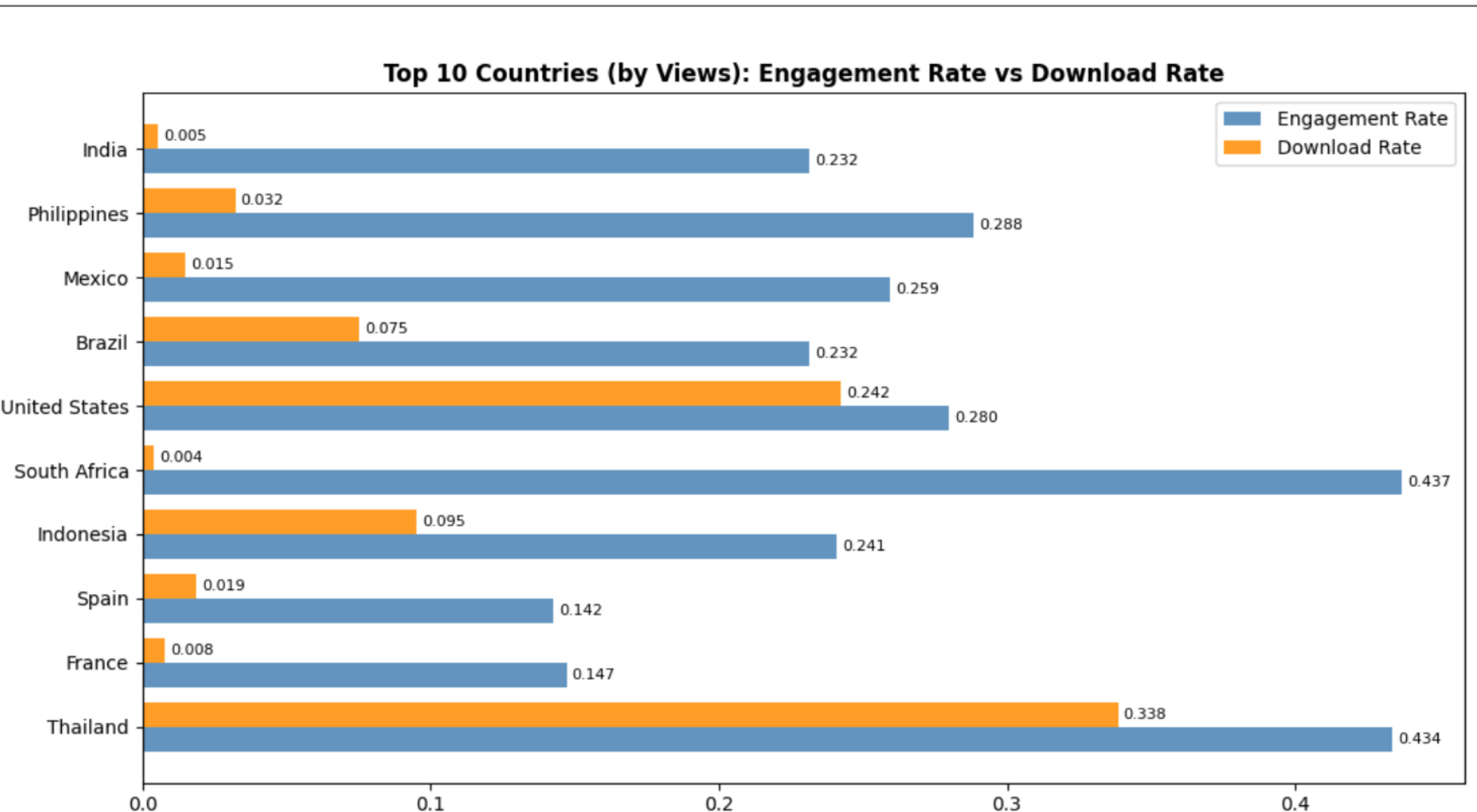
Audience Behavior (Age × Geography × Traffic Source)



Younger users (18-34) show high engagement, but conversion to downloads varies. The 45-54 group have stronger conversion efficiency despite lower reach, indicating higher intent.



Google dominates all three - the primary acquisition channel. Email / Direct traffic shows strong conversion efficiency; social platforms generate reach but lower conversion.



Thailand has high engagement and high download rates despite lower visibility. High-traffic countries like India show lower conversion efficiency.

RECOMMENDATIONS + STRATEGY

Content, Theme

- Prioritize **Education** and **Youth** themes
- Use **video, carousel and image formats** for higher engagement
- Simplify complex topics with **visual storytelling**

Platform

- **YouTube:** expand video-based content
- **LinkedIn:** focus on educational and community organic posts
- **Instagram:** leverage images, carousels and improve reels engagement

Conversion

- Strengthen **calls-to-action** in posts, better content landing page
- Link content to **downloadable resources**
- Optimize for **high-intent traffic (Google, Email)**

Geography

- Increase reach in **Thailand**, use **better content strategies**
- Better reach strategy through Instagram and LinkedIn targeting India and Philippines
- Improve conversion in **high-traffic countries**

CONCLUSION

Content performance depends on aligning **theme, format, and platform**, while audience behavior shows that **high engagement does not always lead to conversion**. Maximizing impact requires balancing **reach and high-intent user actions** across platforms and regions.